

GNU nano 8.7.1

task2.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("Enter command (or exit): ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = 0;

        if(strcmp(command,"exit")==0)
            break;

        pid_t pid = fork();

        if(pid < 0)
        {
            printf("Fork Failed\n");
        }
        else if(pid == 0)
        {
            execlp(command, command, NULL);

            printf("Invalid Command\n");
            exit(1);
        }
        else
        {
            wait(NULL);
        }
    }

    return 0;
```

shreyaj@LAPTOP-FT6VVM4D:~/OSSP\$ tree

```
├── FileManagerment
│   ├── b.txt
│   ├── file1.txt
│   ├── hello
│   ├── newcopy.txt
│   └── numbers.txt
├── LICENSE
├── Makefile
├── ProcessTask
│   └── task2
│       └── task2.c
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── builtin.h
│   ├── design.md
│   ├── executor.h
│   ├── parser.h
│   ├── report.pdf
│   └── shell.h
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
├── structure.txt
└── tests
    ├── test_executor.c
    └── test_parser.c
```

11 directories, 30 files

shreyaj@LAPTOP-FT6VVM4D:~/OSSP\$

```
shreyaj@LAPTOP-FT6VVM4D: ~$ mkdir ProcessTask
mkdir: ProcessTask: File exists
shreyaj@LAPTOP-FT6VVM4D: ~$ cd ProcessTask
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ nano task2.c
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ nano task2.c
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ gcc task2.c -o task2
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ sudo apt update
sudo apt install gcc
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.4 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [321 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.2 kB]
Hit:6 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:8 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Fetched 1118 kB in 5s (208 kB/s)
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
gcc is already the newest version (4:15.2.0-5ubuntu1).
gcc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 99
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$ ./task2
Enter command (or exit): date
Wed Aug  5 09:29:00 UTC 2026
Enter command (or exit): pwd
/home/shreyaj/ProcessTask
Enter command (or exit): ls
task2  task2.c
Enter command (or exit): whoami
shreyaj
Enter command (or exit): exit
shreyaj@LAPTOP-FT6VVM4D: ~/ProcessTask$
```